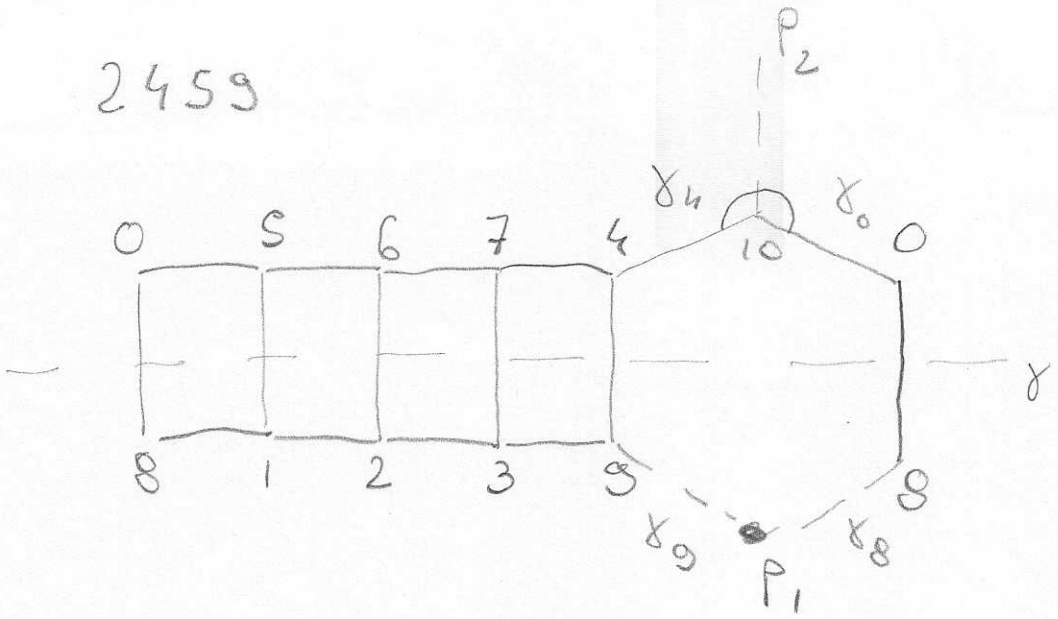
2459



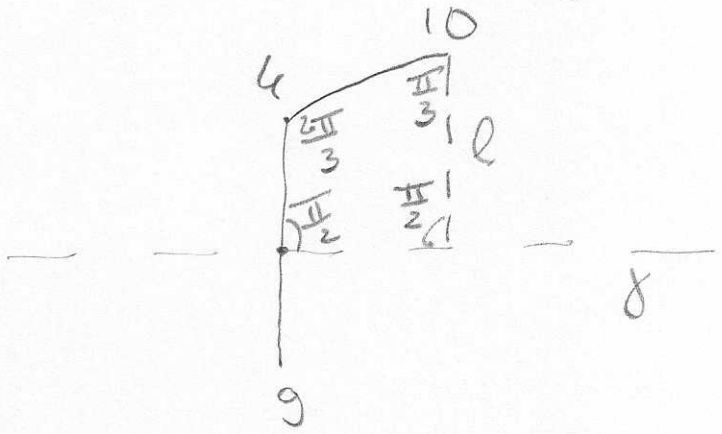
Let γ_4, γ_0, γ_9, and γ_8 be the bisectors of the angles in 4, 0, 9, and 8 and γ be the equator going through the midpoints of the edges 08, 51, 62, 73, and 49.

The symmetry s with respect to γ maps γ_4 in γ_9 and γ_0 in γ_8. So γ_9 and γ_8 meet transversally. Since one intersection is a pin, its antipodal is the other pin and the other intersection of γ_9 and γ_8.

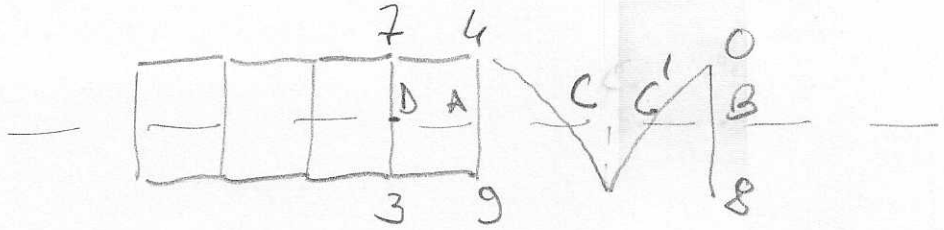
So 10 gets mapped by s either in P_1 or P_2

Case 1 $s(10) = P_2$. Note that 10 is not antipodal to P_2, otherwise it would be P_1.
Then the great circle going through 10 and P_2 is orthogonal to γ.

Therefore 10 cannot be in the same half sphere as 4 and 0, because the quadrilateral formed by 4, 10, A and B would have angles $\frac{2\pi}{3}, \frac{\pi}{3}, \frac{\pi}{2}, \frac{\pi}{2}$



The area of the quadrilateral would need to be 0.
So either 10 is in γ or in the half sphere of 9 and γ



If we let D be the midpoint of 73, A the midpoint of 49, B the midpoint of 08, and C, C' the intersections of 4'10' and 0'10' with BC' and CA are both strictly longer than DA.

But DA is longer than $\frac{\pi}{3}$. So the equator γ would need to be longer than $6 \cdot \frac{\pi}{3}$

Case 2 $s(10) = P_1$

So P, 10 is orthogonal to γ.

As in Case 1 10 must be in the same hemisphere as 4 and 0, while P, is the same as 9 and 8. But then 4'10'0 8P, 9 is an hexagon with angles $\frac{2\pi}{3}$ which does not exist.